IOWA STATE UNIVERSITY

Knowledge-Grounded 3D Disease Editing: What Does 3D Add Beyond 2D?

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Pilot results and three focused tests of what 3D adds
Advisor update

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The Question We Have to Answer

What does knowledge-guided 3D give us that the original 2D photo, or a knowledge-guided 2D edit, does not?

01 **Not enough: 2D → 3D → 2D**
Rendering a reconstruction back to the input view adds nothing the photo did not have.

02 **Candidate gains**
Shape symptoms (curl, pucker, galls), hidden surfaces (leaf underside), views a camera never took, one consistent object across views.

03 **Our rule**
Every claim must be measured, and the measurement must say whether it is *real*, *synthetic* or *unverified*.

How this deck is organised

Act I. Question, five hypotheses, pipeline, status.
Act II. Protocol: audit, case selection, data tiers, structured KB, matched arms.
Act III. Experiments E1–E5, three focused tests of what 3D adds, ablations, statistics, failures.
Act IV. Go / no-go, answers to the 12 questions, next steps.

Status. Pilot runs complete on all 15 specimens; three focused tests added. Numbers carry their *n* and are regenerated from the result files (2026-10-14 17:09).

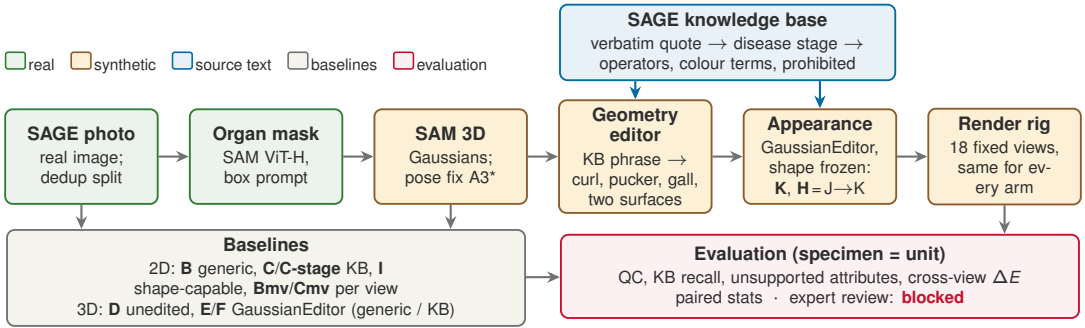
Five Hypotheses and Where Each Stands

	Hypothesis	Evidence so far (specimen = unit)	Status
H1	Single-image reconstruction preserves real anatomy	Input view: SAM 3D pose is off; a pose-only fit lifts median IoU 0.63→0.92 (15/15). Unseen views: *no real reference*.	● input view only
H2	KB improves symptom fidelity over generic prompts	2D colour recall: KB 0.71 vs generic 0.38. 3D shape: KB operators 0.44 vs generic 0.07 (n=6). GaussianEditor: no KB shape at all.	● supported, small *n*, no expert
H3	3D editing is more multi-view consistent than 2D editing	Same-point colour difference: per-view 2D 16.8–19.3 vs 3D 9.3–10.4 (floor 2.0).	✓ internal only
H4	3D holds verifiable information missing from the input view	3D can *represent* surface, direction and volume (Tests 1–3); no public real views to verify them; capture protocol written.	NOT RUN / BLOCKED
H5	KB 3D assets improve a downstream task	Minimal real-photo test (5 crops): no gain over unedited renders; below real-only at $k \leq 4$.	✗ not supported

Bottom line. Supported: a *controllable, auditable KB→shape editor* with consistent views. Not shown: that the 3D content is *true* (H4) or *useful* (H5).

OUR RUN – MEASURED results/alignment/summary.json, results/statistics/{comparisons.json, metrics2d_summary.json, crossview_corr_*.json}. Values regenerated 2026-10-14 17:09.

Pipeline as Implemented



Two editors, not one. GaussianEditor (text → diffusion → Gaussians) changes colour but not shape (E2). Shape therefore comes from our *geometry editor*: each KB term with a defined operator moves the Gaussians coherently; terms without one are logged, never approximated.

OUR RUN – MEASURED diagram traced from the code paths (src/figures/fig1_pipeline.py lists them). SAM 3D Objects, GaussianEditor (Chen *et al.*, CVPR 2024), InstructPix2Pix (Brooks *et al.*, CVPR 2023).

Status at a Glance

15

pilot specimens, 5 SAGE classes, chosen before any model was run

15/15

SAM 3D reconstructions, pose-corrected

9/15

specimens with KB and generic geometry edits (×3 seeds)

Arm	specimens done	state
2D KB-stage / shape edits (C-stage, I)	15/15	complete, 3 seeds
GaussianEditor generic (E)	15/15	complete
GaussianEditor + KB (F)	15/15	complete
KB colour, geometry frozen (K)	15/15	complete
Geometry: generic G / KB J	9/15	done where operator exists

Gate respected. Scale-up waited until Phase A (Peach 0087, every arm) showed coherent shape edits. Phase B finished on two 12 GB TITAN X (about 38 min per GaussianEditor run); the v2 tests take seconds per edit.

OUR RUN – MEASURED counts of asset files under results/pilot and results/phaseA at 2026-10-14 17:09; geometry arms exist only where the KB names an implemented operator: Peach (distort, pucker, thicken), Corn (outgrowth), Mango (depress; its detector is excluded as non-specific). Tomato and Citrus have none.

Phase 0: What the Repository Contained

As found (verified against files and logs)

Claimed	Found
100–150 examples	✗ **one** mango-leaf screenshot
GaussianEditor edits	✗ never ran (env build failed)
SAM 3D needs an A100	✗ runs on 12 GB TITAN X with stage-wise offload (peak 11.6 GB)
3D adds information	✗ that leaf is near-planar (z-extent 4% of length)

Bugs found and fixed while running

01 **KB arm F never ran**
The colon in every KB prompt was parsed as a config key. Prompts now go through validated config files; the prompt the model receives is checked (10 unit tests).

02 **“No visible change” on Peach 0006**
SAM 3D writes infinite opacities → NaN loss → every fp16 step skipped. Sanitised before editing.

03 **GPU 1 crashed at start**
An allocator setting inherited from SAM 3D is unsupported by GaussianEditor’s torch 2.0.1.

04 **Per-view 2D baseline unmasked**
IP2P invented new scenes; re-scored with the object mask.

OUR RUN – MEASURED reports/phase1_protocol/00_repository_audit.md (as-found state), reports/00_full_failure_audit.md (each bug: location, cause, evidence, fix, test).

Phase 1: Cases Where 3D Could Matter

Case	SAGE class	KB text (verbatim source)	What the photo lacks
A geometry	Peach Leaf Curl	"leaves become severely distorted (thickened and puckered)"	pucker depth, far side of curl
B occlusion	Tomato Leaf Mold	"velvety mold forms on the **lower** leaf surface"	underside never photographed
C volumetric	Corn Smut	"tumor-like galls … less than 1 cm to more than 30 cm"	gall volume, back of the ear
D relief	Citrus Canker	"corky with a crater-like depression"	relief under one light
Z control	Mango Anthracnose	flat dark spots on a near-planar leaf	nothing material

How they were chosen

Regex screen of 635 KB classes for 7 shape/occlusion families → read the source clause → inspect images → disease type from SAGE. 3 specimens per class, picked by fixed criteria *before* any model ran.

Not in SAGE or deferred

Tea tortrix, root-knot nematode: pests, absent. Grape Leafroll: canopy shots only. Black Knot: 40 images. Squash Downy Mildew: label-noise flag.

KB TEXT – VERBATIM SOURCE quotes from SAGE KB source pages (Ohioline PLPATH-FRU-26, UMN Extension, APSnet, CDFA). data/metadata/candidate_cases.csv; reports/phase1_protocol/01_case_selection.md.

Phase 2: Ground-Truth Tiers

Tier 1: real multi-view

Same specimen, several real views; one input, rest held out.

Status: none. Every SAGE image is a single, independent view with no specimen ID.

NOT RUN / BLOCKED

Tier 2: measured geometry

Scans or hand-measured traits (curl angle, gall size).

Status: none available.

NOT RUN / BLOCKED

Tier 3: public single view

SAGE: 820,502 image rows, 635 classes with source-grounded KB text.

Status: in use, 15 pilot specimens.

REAL SAGE DATA

Consequence. Without Tier 1/2, every novel view and hidden surface is *synthetic and unverified*. H1 (unseen views) and H4 cannot be tested; we measure fidelity to the input view, KB adherence and internal consistency, and label everything else.

Leakage rules in place: content-hash deduplication before any split (16,973 duplicate rows found); masks from the input image only; no held-out image enters reconstruction, prompting or editing.

REAL SAGE DATA data/metadata/sage_shard_manifest.parquet (all 48 shards); reports/phase1_protocol/02_dataset_protocol.md.

Phase 3: Structured, Stage-Aware KB

Real record: Peach Leaf Curl, stage 0 of 3

```
"source_quote":  "Developing leaves become severely
distorted (thickened and puckered), and have a reddish or
purple cast.  Later, ...  powdery gray ..."
"evidence_status":  "source_supported_unobserved"
"geometry":  [distort ← "severely distorted",
              pucker, thicken ← "thickened and puckered"]
"appearance":  [reddish, purple ← "reddish or purple cast"]
"prohibited":  [stage 1 "powdery gray",
              stage 2 "turn yellow or brown and drop",
              "no geometry operator without a cited phrase",
              "preserve background and unaffected organs"]
```

01 One stage per edit

Mixing stages (red *and* powdery grey) was a bug in the interim prompts; later stages become prohibited edits.

02 Every edit cites a phrase

Operators without a cited phrase are refused; terms without an operator (curl, roll, “corky”) are logged as unsupported.

03 Evidence level kept

Observed in the photo / source-supported but unobserved / hedged / unsupported – never merged.

04 Matched prompts

generic, joint, appearance-only, geometry-only, wrong-stage, other-disease.

KB TEXT – VERBATIM SOURCE data/kb/structured_specs.json (generated by src/kb_grounding/structured.py from the SAGE KB); source: ohioline.osu.edu/factsheet/plpath-fru-26.

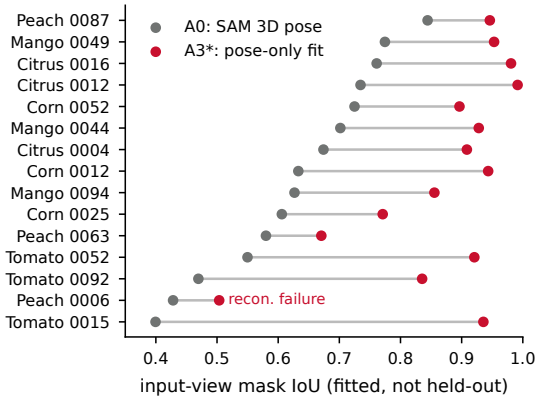
Phase 4: Matched Experimental Arms

Arm	Method	shape?	views	KB	Tests
A	real photo	–	1	–	observed information
B / C	IP2P 2D edit inside the organ mask	no	1	none / KB	is knowledge alone enough?
C-NG / C-MM	2D, KB wrong stage / other disease	no	1	control	negative controls
I	2D, dilated mask (can change outline)	yes	1	KB stage	fairest 2D shape baseline
Bmv/Cmv	IP2P on each rendered view independently	no	8	none / KB	per-view 2D consistency
D	SAM 3D, no edit	–	18	–	reconstruction alone
E / F	GaussianEditor, generic / KB prompt	in principle	18	none / KB	text-driven 3D editing
G / J	geometry editor, random field / KB operators	yes	18	none / KB	does KB control shape?
K / H	GaussianEditor, geometry frozen / J then K	K no, H yes	18	KB	colour only; full method

Fairness controls. 2D and 3D arms receive the *same* KB text. G's random field is matched to J's displacement magnitude (RMS), so only the *direction* of change differs. All 3D arms use one fixed 18-view rig and one renderer. Seeds: 3 for geometry and 2D arms; 1 so far for GaussianEditor arms (38 min each).

OUR RUN – MEASURED reports/05_baseline_comparison.md; experiments/phaseA/run_geometry.py, run_ge_arms.py; src/editing2d/ip2p_batch.py.

E1: The Error Is the Pose, Not the Camera



0.63 → 0.92

median input-view IoU (15/15 improved)

01 Camera checked first
Intrinsics fit the depth map to <2 px (12/15); recentring changes IoU ≤ 0.01.

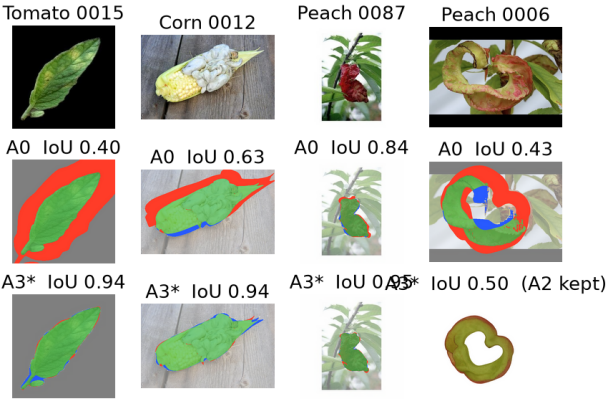
02 Objects come out too large
13/15 oversized (up to +150%); SAM 3D's pose refinement is off by default.

03 Fix touches pose only
Scale/shift, then 7-DoF render-and-compare, kept only if IoU improves (A3 alone diverged on 3/15).

04 Held-out checks are mixed
Depth agreement worse on 6/15; LPIPS worse on 1/15.

OUR RUN – MEASURED results/alignment/summary.json; reports/01_camera_alignment_fixes.md. IoU is fitted to the very mask it is scored on: a fit, not accuracy. Failure threshold (0.6) chosen after seeing the data.

E1: Examples, Including a Real Failure



What works

Leaves, fruit and ears are recovered as one object; after the pose fix the outline matches the photo closely (Tomato 0015: 0.40 $\rightarrow$ 0.94).

What fails

Peach 0006, a strongly curled leaf, becomes a **closed ring**. That is a topology error; no pose change fixes it (best IoU 0.50). Kept in every table.

What we cannot say

Whether the back or underside is right. No second real view exists.

REAL SAGE DATA photos (top row) **OUR RUN – MEASURED** overlays: green = render and mask agree, red = render only, blue = mask only. results/alignment/<specimen>/.

E2: GaussianEditor Changes Colour, Not Shape



All panels SYNTHETIC renders on the same fixed rig; nothing here is verified against a real view.

E/F: jitter, not deformation

Displacement coherence 0.31 / 0.31 (independent noise = 0.25); KB geometric recall 0.

Texture cost

Grain $\times$13.6 (E), $\times$17.0 (F) vs unedited; leaf about twice as thick.

Geometry editor (G, J)

Coherence 0.97 / 0.92; grain $\times$1.1 / $\times$1.2. J follows the KB.

SYNTHETIC – UNVERIFIED Peach Leaf Curl 0087, all arms, same 18-view rig. OUR RUN – MEASURED medians over specimens: E n=15, F n=15, G n=9, J n=9. Prune-aware Gaussian matching: results/phaseA/*/<arm>/seed*/gaussian_diff.json.

E2: KB-Driven Shape Edits on Gaussians

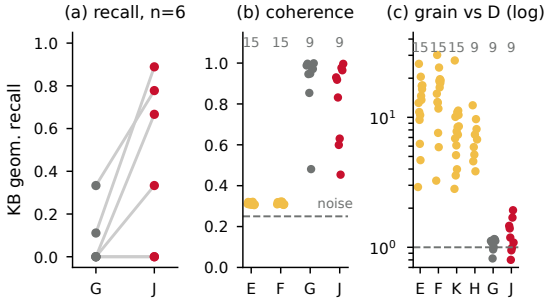
Method

$\mu_i' = \mu_i + m_i\, d_i$

$\mathcal{L} = \mathcal{L}_{\text{edit}} + 0.1\,\mathcal{L}_{\text{ARAP}} + 0.01\,\mathcal{L}_{\text{smooth}} + \mathcal{L}_{\text{anchor}}$

KB term → operator → target field h on the surface (8-NN graph). Rotations follow the new normals.

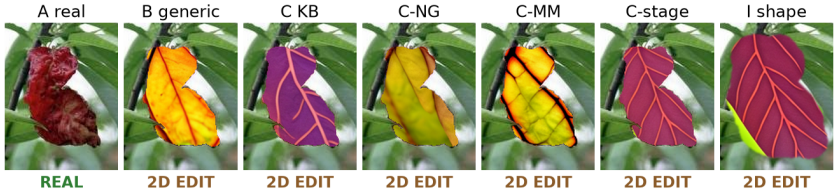
KB term	operator
puckered	Poisson-disk blisters
distorted	low-frequency warp
thickened	thicker along own normal
galls, tumors	outgrowths
sunken	depressions
curl, roll, corky	✗ none yet (logged)

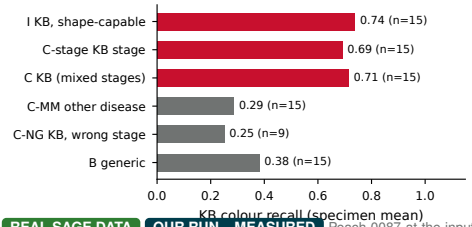


KB vs generic, equal magnitude: geometric recall 0.44 vs 0.07 (+0.37; 95% CI +0.13 to +0.61; J better on 4/6, worse on 0). Only 2 classes, and partly by construction: J aims at the attributes the detector measures.

OUR RUN – MEASURED results/statistics/comparisons.json, specimen_metrics.csv; src/geometry_editor/deform.py. Detector thresholds set on a synthetic sheet; blister count → density change is **POST-HOC CHANGE (LOGGED)**.

E2: Knowledge Steers Colour in 2D Too





Knowledge, not 3D, drives colour

KB colour recall 0.71 (C) and 0.69 (C-stage) vs 0.38 generic; wrong-stage and other-disease controls fall to 0.25 / 0.29. Generic edits drift to “autumn” yellow.

2D can change shape, but not correctly

Arm I changes the outline (outside-mask change 0.83) yet *replaces* the real puckering with a flat, smooth leaf.

REAL SAGE DATA | OUR RUN – MEASURED Peach 0087 at the input view: real photo and 2D arms. results/statistics/metrics2d_summary.json (n per arm on the chart).

Colour detectors: Lab shifts, $T = 5$. “Purple” detector and “pale” definition are POST-HOC CHANGE (LOGGED); both versions in report 05.

E3: What Does the 3D Asset Add? Three Focused Tests

01 Direction
Can a cited direction (“curl upwards”) be imposed and measured, with the opposite direction as a control?

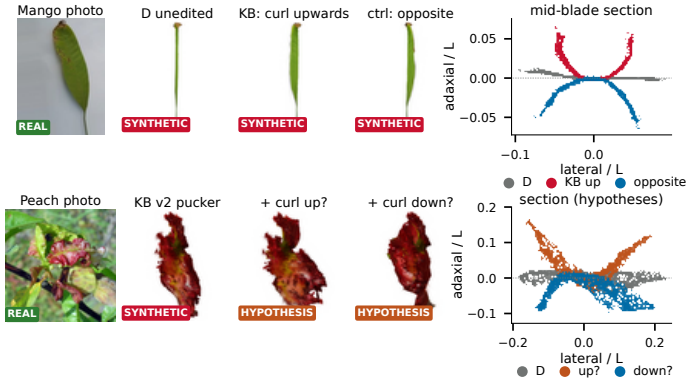
02 Volume
Can a gall be a closed volume that replaces kernels, where a height field only pushes the surface out?

03 Surface
Can a lower-surface symptom exist without appearing on the upper surface, which is all a photo shows?

Design. Smallest tests that can fail: 3 specimens per disease, 3 seeds, every method rendered from the same 18 cameras and the corrected input camera, each with a placement or direction control. Original assets are never modified.

OUR RUN – MEASURED experiments/demos/run_demos.py; results/demos/execution_log.jsonl. Tests follow on the next three slides.

Test 1: Directional Shape From the Text



Direction follows the cited phrase

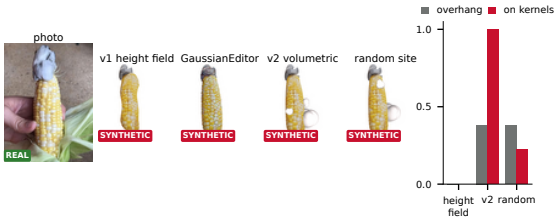
Mango, “Leaves curl upwards”: margins rise in 3/3 leaves (4.8° →46.2°); reversed control falls in 3/3 (-35.4°). Strain p95 11% vs 7% generic.

Peach: the source gives no direction

So the operator **abstains**; up/down appear only as labelled hypotheses. Pucker placed on reddish tissue: enrichment 1.14 vs 1.07 uniform, no localisation gain.

SYNTHETIC – UNVERIFIED **OUR RUN – MEASURED** results/demos/{mango_curl,peach}/*/metrics.json; roll = signed margin angle about the midrib (+ = toward the upper side). The Mango specimens have anthracnose: the curl is a counterfactual, and it lowers input-view IoU (0.91→0.62), i.e. it contradicts the flat leaf in the photo.

Test 2: Galls as Volumes on Kernel Tissue



What a height field cannot do

The v1 operator only pushes the surface out: no underside, no overhang. v2 galls are closed lobed bodies; 0.38 of their surface overhangs the ear.

Placement from evidence

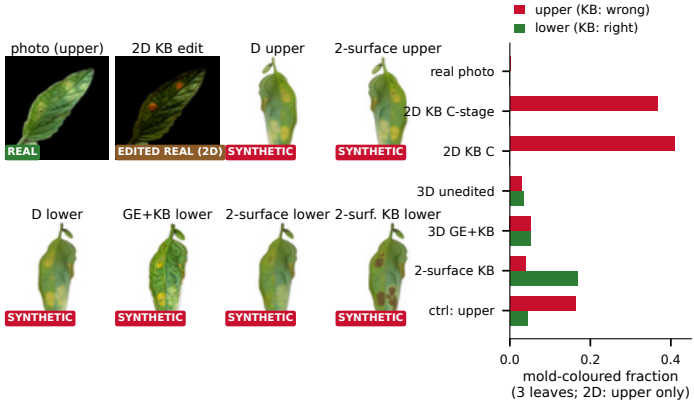
"...develop on the ear replacing kernels" (UNL): galls start on kernel tissue in 1.00 of runs vs 0.22 for random sites; replaced kernels are hidden.

What is assumed

Ear length 18 cm (no ruler) for the 1–30 cm range; count and along-ear position are not in any source. 1 ear abstained: no kernel tissue identifiable.

REAL SAGE DATA | SYNTHETIC – UNVERIFIED | OUR RUN – MEASURED results/demos/corn/*/metrics.json, n = 2 ears × 3 seeds. Mean gall volume 35.7 cm^3 (assumed scale) vs 26.4 cm^3 added by the height field. Volume and overhang are properties of the edit, not a validation against real galls.

Test 3: Upper vs Lower Leaf Surface



2D puts the mold on the wrong side

KB: mold "on the lower leaf surface below leaf spots". A photo shows the upper side only; KB 2D edits paint mold colour on 17–49% (C-stage) / 31–53% (C) of it; real: 0–0%.

Two surfaces keep it apart

One sheet mirrors upper spots below (12–20% of lower pixels). Two surfaces: mold 7–29% of lower pixels, upper 0–8% (baseline 0–7%); 73–84% of it under real spots (random 7–23%).

REAL SAGE DATA | SYNTHETIC – UNVERIFIED | OUR RUN – MEASURED — results/demos/tomato/*/metrics.json, 3 leaves × 3 seeds, all 18 rig views. Colour classes fixed before the edits. The real underside is *unknown*: this shows the KB statement can be represented, not that it is right.

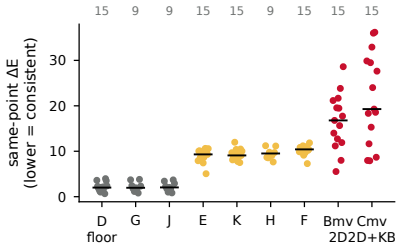
Answer: What Does KB-Guided 3D Add Beyond 2D?

Capability	Measured here	Status
Which *surface* a symptom is on	2D KB edits: mold on the visible upper side (17–49%); two-surface 3D: lower side only	✓ representable, measured
Which *direction* a leaf deforms	cited direction reproduced 3/3, reversed control 3/3; Peach abstains	✓ when the text states it
Symptom *volume* and overhang	closed galls on kernel tissue (1.00); height field: no overhang	✓ representable
One object across views	same-point ΔE 3D 2.0–10.4 vs per-view 2D 16.8–19.3	✓ internal only
Hidden symptoms are *correct*	no matched real views (search: none public)	NOT RUN / BLOCKED
Biologically realistic	no ratings yet (package ready)	NOT RUN / BLOCKED
Better downstream classifier	minimal test: NOT SUPPORTED (falsified at this scale)	✗ not supported

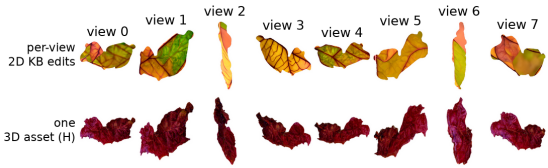
Direct answer. KB-guided 3D can *express* source-grounded facts that one image cannot: the surface, the direction and the volume of a symptom, consistently across views and without contradicting the photo. That capability is measured. Whether the expressed shapes are *true* for the specimen, or *useful* for a real task, is not shown yet.

OUR RUN – MEASURED NOT RUN / BLOCKED results/demos/summary.json; results/downstream_minimal/summary.json; reports/11–14 in the SAGE-Extend repository.

E4: One 3D Asset vs Per-View 2D Edits



Metric. The same surface point is projected into adjacent views (depth-tested); we report its colour difference ΔE. D (unedited) gives the resampling floor.

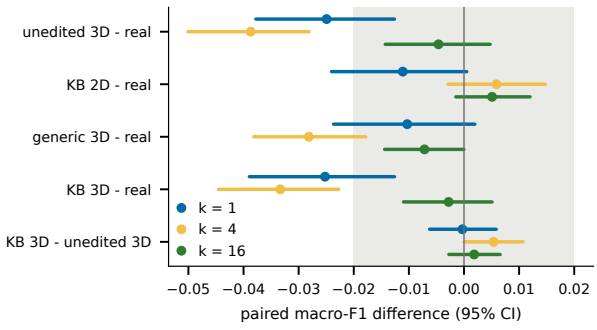


Per-view 2D: 16.8 (generic), 19.3 (KB). 3D GaussianEditor: 9.3–10.4. Geometry editor: 2.0, the floor 2.0.

Consistency is not accuracy. A leaf can be perfectly consistent and biologically wrong: the 0006 ring is consistent in all 18 views.

SYNTHETIC – UNVERIFIED OUR RUN – MEASURED results/statistics/crossview_corr_*.json; n per arm on the chart (Bmv/Cmv on the 4 Peach/Tomato specimens with masked per-view edits).

E5: Minimal Downstream Test on Real Photos



k real/class	KB-3D − real only	KB-3D − unedited 3D
1	-0.025	-0.000
4	-0.033	+0.005
16	-0.003	+0.002

Verdict (pre-registered): NOT SUPPORTED (falsified at this scale). The knowledge edit adds under 2 macro-F1 points over unedited renders; renders themselves hurt at small *k*. The render-to-photo gap comes before knowledge.

OUR RUN – MEASURED reports/14_downstream_minimal.md: within-crop disease classification, 5 crops, frozen DINOv2 + linear probe, 50 paired draws per *k*, 2,936 held-out real SAGE photos, pHash-disjoint. Macro-F1 differences; 95% CIs in results/downstream_minimal/contrasts.csv.

Validation: Raters and Real Multi-View Data

Expert review package (built, not yet sent)

84 blinded, shuffled items (72 test, 9 repeats, 3 catch); real photo beside each candidate; verbatim KB text; 6 anchored 1–5 criteria incl. surface and shape. About 57 min per rater. Agreement code (Krippendorff α, weighted κ, ICC) passes 14/14 tests against textbook values.

Limit

2D vs 3D cannot be fully blinded: one photo and a multi-view strip look different.

Matched multi-view data

Search of about 30 public datasets: **none** pairs both leaf surfaces, 3D curl, or gall volume for a diseased specimen. Protocol: 30 specimens (8 per disease + controls), 98 views each incl. an underside ring, flat photos of both surfaces, one view as input, the rest held out.

Timing

Corn smut: July–Sept.; Tomato Leaf Mold: greenhouse; Peach Leaf Curl: April–May 2027. About 45 person-hours.

NOT RUN / BLOCKED reports/11_multiview_data_search.md, 12_acquisition_protocol.md, 13_expert_review_protocol.md; results/review_package_v1/ (key file kept separately).

Ablations: Knowledge × Geometry

2×2 on total KB recall

	no KB	KB
no geometry	D 0.00	K 0.17
geometry	G 0.04	H 0.53

Interaction $(H-G)-(K-D) = +0.32$, **n = 6** specimens with every cell (2 classes).

Ablation	Arms	Status
No KB / generic	B, E, G	✓
KB, wrong stage	C-NG	✓
Mismatched KB (neg. ctrl)	C-MM	✓
KB appearance only	K	✓
KB geometry only	J	✓
Appearance vs geometry 3D	K vs J vs H	✓
Placement: evidence vs random	J2/J2u, V/Vr, BLk/BLr	✓
Direction: cited vs opposite	CU/CD	✓
Viewpoints (1/2/4/8/adaptive)	–	✗
Severity, occlusion, background	–	✗

OUR RUN – MEASURED results/statistics/ablation_2x2.json. Total KB recall = mean of geometric and colour recall over the stage's required attributes.

Statistics: Specimen Is the Unit

Comparison	n (classes)	means	Δ [95% CI]	Holm p	Reading
KB geometric recall, J vs G	6 (2)	0.44 / 0.07	+0.37 [+0.13, +0.61]	0.88	direction only, 2 classes
Unsupported attributes, J vs G	9 (3)	0.06 / 0.17	[-0.28, +0.03]	1.00	inconclusive
Coherence, J vs E	9 (3)	0.81 / 0.31	[+0.37, +0.61]	0.043	survives Holm
Texture grain, J vs E	9 (3)	1.28 / 12.94	[-15.89, -7.94]	0.043	survives Holm
Texture grain, K vs E	15 (5)	8.71 / 16.98	[-14.15, -4.24]	0.001	survives Holm
Texture grain, H vs F	9 (3)	7.09 / 15.73	[-11.96, -5.45]	0.043	survives Holm

Procedure

Seeds averaged within specimen; paired Wilcoxon; Holm correction across comparisons; specimen bootstrap CI and class-cluster bootstrap (flagged when <3 classes).

What this means

Artefact measures (coherence, grain) now survive Holm with 9–15 specimens. KB recall does not: $n \leq 9$, at most 3 classes. These are editor properties, not biological validity.

OUR RUN – MEASURED results/statistics/comparisons.json; reports/07_statistical_analysis.md. CIs: specimen bootstrap, 10,000 resamples.

Compute: Two 12 GB TITAN X GPUs

Step	median wall-clock	runs ok / tried
SAM 3D reconstruction (offloaded)	1070 s	15 / 15
Pose alignment A0–A3	78 s	15 / 15
Geometry edit J or G (+18 renders)	44–45 s	54 / 54
IP2P 2D edit, GPU / CPU	58 / 238 s	158 / 158
GaussianEditor, 200 steps	2229–2251 s	11 / 17

Peak VRAM: SAM 3D 11.6 GB of 12. All 6 failed GaussianEditor attempts were the two bugs on slide “Phase 0” and were rerun after the fix.

≈3.4 GPU-h

per specimen for the full arm set

≈50×

GaussianEditor vs geometry editor run time

(2251 s vs 44 s per edit)

The cheap editor is also the one that produces the shape change: cost is not the bottleneck for scaling J/H.

OUR RUN – MEASURED reports/08_gpu_compute_audit.md, generated from JSONL run logs and nvidia-smi samples (results/compute/gpu_samples.csv).

Failure Taxonomy

Peach 0006 photo (curled leaf)



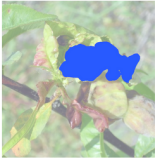
REAL

SAM 3D: closed ring



SYNTHETIC

pose fit A3 diverged (0063, IoU 0.00)



GaussianEditor: grain + jitter



SYNTHETIC

2D shape edit: puckering flattened



EDITED REAL (2D)

Reconstruction

Topology error (ring); 13/15 over-scaled; leaves are single sheets with view-independent colour, so upper and lower surfaces cannot differ.

Optimisation

Pose refinement diverged on 3/15; needs the accept-if-better rule.

Editing

GaussianEditor: grain and jitter, no shape. J: blisters placed randomly, not at observed lesions.

Biology

2D shape edit removes real puckering; generic edits make “autumn” colours; stage mixing in old prompts.

REAL SAGE DATA | SYNTHETIC – UNVERIFIED reports/09_failure_analysis.md; 00_full_failure_audit.md (root cause and status of each).

Go / No-Go After the Pilot

Decision	Required evidence vs what we have	Verdict
GO: KB-grounded 3D reconstruction	Verified hidden-trait recovery and a measured gain over KB 2D. *No held-out real views exist; minimal downstream test: no gain.*	✗ not met
PIVOT: 3D simulation and inspection	Plausible, controllable shape edits with limited claims. *Surface, direction and volume are representable and measured; consistent views; not yet expert-rated.*	● **best fit**
PIVOT: 2D knowledge-guided editing	Knowledge helps, 3D adds nothing. *Knowledge does steer 2D colour; but 2D cannot represent shape correctly (arm I).*	● partly
NO-GO: unreliable synthetic biology	Errors dominate. *Failures are real (ring, GE jitter) but diagnosed, reproducible and avoidable.*	✗ not dominant

Recommendation. Narrow the claim to *knowledge-controlled 3D symptom simulation*: an auditable KB→shape editor whose edits are traceable to cited text and consistent across views. Do not claim reconstruction of hidden anatomy. Numerical go/no-go thresholds will be set from pilot variances, not before.

OUR RUN – MEASURED evidence from slides E1–E4; decision logic from reports/10_final_experimental_protocol.md (Outcome D for now: no established advantage on real data).

The Twelve Questions (1/2)

#	Question	Answer from the evidence
1	Why is 3D necessary here?	To state which surface, which direction and how much volume (Tests 1–3). For flat spots it adds nothing.
2	Which symptoms gain from 3D?	Underside sporulation (Tomato Leaf Mold, 67 SAGE records name a lower surface), galls (Corn Smut), directional curl (42 records name curl/roll; direction is given in some, not for Peach).
3	How reliable is one-image reconstruction?	Input view after pose fix: median IoU 0.92; 1/15 topology failure. Unseen views: unknown.
4	What does the KB contribute?	Which attributes to edit, at which stage, and which edits are prohibited. It raises KB recall in 2D (0.71 vs 0.38) and 3D shape (0.44 vs 0.07).
5	Can it be independently verified?	Text-level yes (cited phrase per edit). Specimen-level no: needs real multi-view data or expert raters.
6	Hidden symptoms recovered or just plausible?	Represented, not recovered. Nothing hidden can be verified with SAGE or any public set we found.

OUR RUN – MEASURED all numbers are macros generated from results/ (2026-10-14 17:09).

The Twelve Questions (2/2)

#	Question	Answer from the evidence
7	Is 3D editing better than KB 2D editing?	For colour: no measured advantage. For shape: 2D either cannot change it (mask) or changes it wrongly (arm I); 3D J/H can. For consistency: 3D far better (2.0–9.3 vs 16.8–19.3).
8	Knowledge vs geometry share of the gain?	2×2 on 6 specimens: interaction +0.32; each factor adds a different attribute type.
9	Does it improve a real agricultural task?	Minimal test on real photos: not supported (KB-3D vs unedited renders $\leq$ +0.02 macro-F1). Full E5 not justified yet.
10	Strongest failure modes?	Topology failure on curled leaves; GaussianEditor jitter and grain; render-to-photo gap; edits that contradict a flat photo (curl lowers input IoU); no ruler for physical size.
11	Experiments still required?	Send the expert package; capture 30 multi-view specimens (protocol ready); close the render gap; then E5.
12	Ready for publication?	Not yet as "reconstruction". A narrower paper (auditable KB$\rightarrow$3D symptom editing, with honest limits) is within reach after expert review.

OUR RUN – MEASURED **NOT RUN / BLOCKED** item 9: reports/14; item 11 depends on data or raters we do not yet have.

Next Steps, in Priority Order

01 Send the expert package
84 blinded items, 3 raters, about 1 h each; agreement analysis is coded and tested. The only route to biological validity now.

02 Capture real multi-view specimens
Corn smut and Tomato Leaf Mold first (available now); Peach Leaf Curl in spring 2027. Held-out views test surface, direction and volume.

03 Close the render-to-photo gap
Renders hurt the classifier before any knowledge is added; fix backgrounds and lighting before E5.

04 Find a source for Peach curl direction
Or keep abstaining; never infer it.

05 Full E5 only if 01–03 pass
The 250 GPU-h study waits for a validated, gap-free asset.

Done since the last update
E/F/K/H on all 15 specimens; directional curl, two-surface leaves, volumetric galls, evidence placement; 3 focused tests; minimal downstream test; expert package; data search and capture protocol.

Question for you. Can we fund about 45 person-hours of capture and recruit three raters? Both are needed before any validity claim.

OUR RUN – MEASURED RESEARCH_PROGRESS.md and reports/11–15 in the SAGE-Extend repository.

Remaining Blockers

Blocker	Effect	Unblocked by
No matched real views	hidden surface, direction and volume cannot be scored	capture protocol (30 specimens)
No expert raters	biological realism unmeasured	package v1, 3 raters
No physical scale in photos	gall size in cm rests on an assumed ear length	ruler in future captures
Peach curl direction not in source	curl abstains for the demo disease	a verbatim source
Render-to-photo gap	synthetic data hurts a real classifier	rendering work before E5
Single-image reconstruction errors	ring topology (Peach 0006), over-scale	multi-view input
Disk on the workstation	13 GB free	archive old runs

NOT RUN / BLOCKED full list with owners: reports/15_revised_scientific_report.md, section “Blockers”.

Deliverables Checklist

Deliverable	built	tested	real data	Blocked by / evidence still needed
Code audit; failure audit	✓	✓	✓	–
Ranked case shortlist	✓	✓	✓	expert check of SAGE labels
Dataset and ground-truth protocol	✓	✓	●	Tier 1/2 data
Structured KB + validator	✓	✓	✓	expert check of operator mapping
Baseline arms A–K, shared runner	✓	✓	✓	–
E1 reconstruction	✓	✓	●	held-out real views
E2 editing fidelity	✓	✓	●	expert ratings
E3 three focused tests (v2 editor)	✓	✓	●	held-out real views
E4 multi-view consistency	✓	✓	●	(accuracy needs real views)
E5 minimal downstream	✓	✓	✓	full study gated
Expert review package	✓	✓	–	raters
Multi-view search + protocol	✓	–	✗	capture
Ablation matrix	✓	●	●	viewpoint, severity ablations
Statistics plan and code	✓	✓	✓	larger *n*
QC, compute logging, resume	✓	✓	✓	–
Figure code from results	✓	✓	✓	–
Revised report (reports/15)	✓	–	–	–

OUR RUN – MEASURED reports/00–10, NEXT_EXPERIMENT_RESULTS.md, RESEARCH_PROGRESS.md in the SAGE-Extend repository. ✓ done ● partial ✗ not done.

Selected References

[1] SAM 3D Team, X. Chen *et al.*, “SAM 3D: 3Dfy Anything in Images,” *arXiv:2511.16624*, 2025.

[2] Y. Chen *et al.*, “GaussianEditor: Swift and Controllable 3D Editing with Gaussian Splatting,” *CVPR*, 2024.

[3] B. Kerbl *et al.*, “3D Gaussian Splatting for Real-Time Radiance Field Rendering,” *ACM TOG*, 2023.

[4] T. Brooks, A. Holynski and A. A. Efros, “InstructPix2Pix: Learning to Follow Image Editing Instructions,” *CVPR*, 2023.

[5] A. Haque *et al.*, “Instruct-NeRF2NeRF: Editing 3D Scenes with Instructions,” *ICCV*, 2023.

[6] O. Sorkine and M. Alexa, “As-Rigid-As-Possible Surface Modeling,” *Symposium on Geometry Processing*, 2007.

[7] Disease text: SAGE KB source pages (Ohioline PLPATH-FRU-26; UMN Extension; APSnet; Plantix) and UNL CropWatch (corn smut), quoted verbatim.

Backup: GaussianEditor (CVPR 2024)



Teaser (Fig. 1) from Chen *et al.*, CVPR 2024; github.com/buaacyw/GaussianEditor.

Semantic tracing. Edits only the target Gaussians.

Hierarchical splatting. Stable under 2D guidance.

Diffusion guidance. InstructPix2Pix edits rendered views from text; Gaussians are fit to them.

Our finding. On SAM 3D leaves it changes colour, not shape (E2).

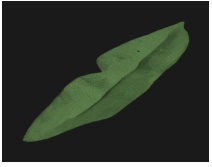
Backup: The Original One-Leaf Prototype



Rule-based lesions



Yellowing, holes



SAM 3D mesh

Where we started

One mango-leaf screenshot, rule-based re-colouring, GaussianEditor not yet running. The leaf is nearly planar, and anthracnose is a surface symptom: a poor test of what 3D adds. This is why Phase 1 re-selected the cases.

Earlier deck (October 2026), figures from github.com/ushashi149/SAGE-Extend (commit 371d96b).